

Exercise 4.2.7. Show that the Minkowski sum $\mathcal{P}_1 + \mathcal{P}_2$ of two convex polytopes $\mathcal{P}_1 \subset \mathbb{R}^d$, $\mathcal{P}_2 \subset \mathbb{R}^d$ is again a convex polytope. In particular, if $\mathcal{P}_i = \text{Conv}(\mathcal{A}_i)$ for a finite set $\mathcal{A}_i \subset \mathbb{R}^d$, $i = 1, 2$, then we have $\mathcal{P}_1 + \mathcal{P}_2 = \text{Conv}(\mathcal{A}_1) + \text{Conv}(\mathcal{A}_2) = \text{Conv}(\mathcal{A}_1 + \mathcal{A}_2)$.

$$\text{Conv}(\mathcal{A}_1) + \text{Conv}(\mathcal{A}_2) = \text{Conv}(\mathcal{A}_1 + \mathcal{A}_2)$$

" $\supset$ " $x \in \text{Conv}(\mathcal{A}_1 + \mathcal{A}_2)$

$$x = \sum_{i,j} d_{ij} (a_i + b_j) \quad a_i \in \mathcal{A}_1, \quad b_j \in \mathcal{A}_2, \\ \sum_{i,j} d_{ij} = 1, \quad d_{ij} \geq 0$$

$$= \underbrace{\sum_{i,j} d_{ij} a_i}_{\in \text{Conv}(\mathcal{A}_1)} + \underbrace{\sum_{i,j} d_{ij} b_j}_{\in \text{Conv}(\mathcal{A}_2)}$$

" $\subset$ " $x \in \text{Conv}(\mathcal{A}_1) \quad y \in \text{Conv}(\mathcal{A}_2)$

$$x = \sum_i c_i a_i, \quad \sum_i c_i = 1, \quad c_i \geq 0$$

$$y = \sum_j d_j b_j, \quad \sum_j d_j = 1, \quad d_j \geq 0$$

$$x + y = \sum_i c_i a_i + \sum_j d_j b_j$$

$$= \left(\sum_j d_j \right) \sum_i c_i a_i + \left(\sum_i c_i \right) \sum_j d_j b_j$$

$$= \sum_{i,j} \underbrace{c_i d_j}_{\geq 0} (a_i + b_j)$$

$$\sum_{i,j} c_i d_j = \sum_i c_i \left(\sum_j d_j \right) \\ = \sum_i c_i = 1$$

Exercise 4.3.6. Show that if $\mathcal{P} = \mathcal{P}_1 = \dots = \mathcal{P}_d$, then $\text{MV}(\mathcal{P}_1, \dots, \mathcal{P}_d) = d! \text{Vol}(\mathcal{P})$.

$$\text{Vol}(w_1 \mathcal{P}_1 + \dots + w_d \mathcal{P}_d)$$

$$\stackrel{\mathcal{P}_i = \mathcal{P}}{=} \text{Vol}((w_1 + \dots + w_d) \mathcal{P})$$

$$= \int_{(w_1 + \dots + w_d) \mathcal{P}} dx_1 \dots dx_d$$

$$= \int_{\mathcal{P}} d(w_1 + \dots + w_d) x_1 \dots d(w_1 + \dots + w_d) x_d$$

$$= (w_1 + \dots + w_d)^d \int_{\mathcal{P}} dx_1 \dots dx_d$$

$$= (w_1 + \dots + w_d)^d \text{Vol}(\mathcal{P})$$

the coefficient of $w_1 \dots w_d$ of $(w_1 + \dots + w_d)^d$
is $\underline{d!}$

Exercise 4.3.8. The matrices $A_1 = \begin{pmatrix} 0 & 1 & 2 \\ 0 & 1 & 2 \end{pmatrix}$ and $A_2 = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 \end{pmatrix}$ give rise to a surface $X_{A_\bullet} = X_{(A_1, A_2)} \subset \mathbb{P}^2 \times \mathbb{P}^3$ (Example 4.1.12). Use the code from Example 4.3.3 to verify that $[X_{A_\bullet}] = 2h_1h_2^2 + 2h_1^2h_2$. Match this with Theorem 4.3.7. [Note that the toric variety $X_{A_1} \subset \mathbb{P}^2$ is the curve given by $x_1x_3 - x_2^2 = 0$, and $\delta_{(2,0)} = 0$ is the number of points on that curve satisfying two additional generic linear equations.

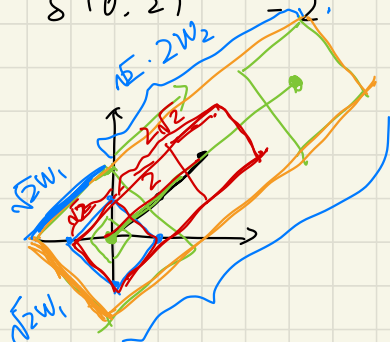
$$S = \begin{bmatrix} 1 & \dots \\ 2 & \dots \end{bmatrix}$$

$$[Z^2 : Z'A_\bullet] = 2 \quad \text{Example 4.1.12}$$

$$\delta_{(2,0)} = 0 \quad \text{MV}(P_1, P_1) = 2! \text{Vol}(P_1) = 0$$

$$\delta_{(1,1)} = 2 \quad \text{MV}(P_1, P_2) = 4$$

$$\delta_{(0,2)} = 2 \quad \textcircled{1} \text{Vol}(w_1P_1 + w_2P_2) = \sqrt{2}w_1(2\sqrt{2}w_2 + \sqrt{2}w_1) = 4w_1w_2 + 2w_1^2$$



$$\textcircled{2} \text{MV}(P_1, P_2)$$

$$\underbrace{\text{Vol}(P_1 + P_2)}_{\substack{6 \\ 3\sqrt{2} \cdot \sqrt{2}}} - \underbrace{\text{Vol}(P_1)}_{0} - \underbrace{\text{Vol}(P_2)}_{2} = 4$$

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# Exercise 4.3.8
A1 = [0 1 2; 0 1 2]
A2 = [-1 1 0 0; 0 0 -1 1]
Cay = [A1 A2; ones(Int, 1, 3) zeros(Int, 1, 4); zeros(Int, 1, 3) ones(Int, 1, 4)]
using Oscar
I = toric_ideal(transpose(Cay))

R, x, y = polynomial_ring(QQ, :x=>1:3, :y=>1:4)
phi = hom(base_ring(I), R, vcat(x, y))
I = phi(I)

for i = 2:-1:0
    J = [(rand(-100:100, 1, 3)*x)[1] for j = 1:i]
    J = [J; [(rand(-100:100, 1, 4)*y)[1] for j = 1:2-i]]
    K = saturation(I + ideal(J), intersect(ideal(x), ideal(y)))
    println("delta_($i,$(2-i)): $(degree(K))")
end

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delta_(2,0): 0
delta_(1,1): 2
delta_(0,2): 2

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Exercise 4.3.17. Compute the multidegree $[X_{A_\bullet}]$ and the Hilbert polynomial $\text{HP}_{X_{A_\bullet}}$ for the multiprojective toric variety defined by $A_1 = \begin{pmatrix} 9 & 7 & 4 & 5 \\ 6 & 0 & 9 & 4 \end{pmatrix}$ and $A_2 = \begin{pmatrix} 6 & 7 & 3 & 2 \\ 6 & 5 & 0 & 8 \end{pmatrix}$.

-- Exercise 4.3.17

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A1 = matrix {{9,7,4,5}, {6,0,9,4}}
A2 = matrix {{6,7,3,2}, {6,5,0,8}}
n = {rank(source A1), rank (source A2)}
toprows = transpose (transpose (A1) || transpose (A2))
r1 = join(apply (n_0, i -> 1), apply(n_1, i -> 0))
r2 = join (apply(n_0, i -> 0), apply(n_1, i -> 1))
bottomrows = matrix{r1,r2}
Cay = toprows || bottomrows

needsPackage "Quasidegrees"
degs = join(apply(n_0, i -> {1,0}), apply(n_1, i -> {0,1}))
S = QQ[X_1..X_(n_0), Y_1..Y_(n_1), Degrees => degs]
IA = toricIdeal(Cay,S)

needsPackage "CorrespondenceScrolls"
M = cokernel matrix gens IA
multiHilbertPolynomial(M)
```

$$o15 = 21s^2 + 53s + \frac{39}{2}s^2 - 85s - \frac{169}{2}s - 63$$